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REALIZATION CALCULUS



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# Circuit Obstructions

## Restoration, complete covers, and the limits of exact projection

Visual guide

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REALIZATION CALCULUS

## Reading scope

Exact examples, finite certificates, and the meaning of the figures

Restoration / Coverage / Projection

## How to read this companion

This guide explains the maintained results through explicit states, finite truth tables, evaluation vectors, and small algebra examples. Every chart is generated by exact Python arithmetic. The figures are illustrations of specified objects; a visual pattern is never promoted to a universal theorem.

The main paper supplies the written proofs. The Lean guide identifies the formal definitions and all 115 declarations in the archived Block 02. This companion connects those statements to examples a reader can recompute.

| Pages | Reading route |
|---|---|
| 2–8 | The Boolean model, arbitrary cuts, dirty scratch, and sharing |
| 9–13 | Complete coverage, affine masks, and exact dual certificates |
| 14–18 | Small calibration and the costs and safety of projection |
| 19–20 | Orthogonal witnesses and the faithful-summary theorem |
| 21–25 | Weighted fingerprints, trace, overlap, and randomized evaluation |
| 26 | Reproduction, evidence labels, and source identity |

Three labels have different meanings. *Written proof* identifies a general argument with stated hypotheses. *Exact finite check* identifies an executed assertion with a bounded range. *Archived formal result* identifies the preserved Lean evidence for its pinned source revision. A fresh formal replay was attempted on 7 October but stopped because the Lean compiler is unavailable. It is not recorded as a new kernel pass.

The fixed scope is C01–C08. No figure establishes unrestricted circuit hardness, an efficient weighted-trace evaluator, or a SAT algorithm. The distinction between nonlinear gate count, total gate count, input length, and summary dimension is part of the mathematics throughout this guide.

**Evidence.** Model-assisted source review and freshly executed exact fixtures; not independent human review. Original source release v1.2.0 is preserved.

## The model: boundary restoration and Boolean degree

A state has logical input wires $x$, zero-initialized ancillary wires, and a target bit $y$. A restored oracle is a legal classical circuit satisfying

$$C(x, 0^a, y) = (x, 0^a, y + f(x)) \qquad \text{for every } x \text{ and both } y.$$

The input and scratch restoration conditions hold at the boundary. Wires may change internally, and the target may control an intermediate gate. Each primitive gate is an involution:

$$y \leftarrow y + 1, \quad y \leftarrow y + x_i, \quad y \leftarrow y + x_i x_j.$$

In a CNOT the control differs from the target. In a Toffoli all three wires are distinct. These are construction rules, not informal assumptions.

All polynomial arithmetic in the Boolean state model is over $\mathbb{F}_2$, with $x_i^2 = x_i$. Every function has a unique multilinear algebraic normal form. Its degree is the largest support size with a nonzero coefficient; the zero function has mathematical degree $-1$ in the Python helpers. A formal degree bound uses a natural number, so zero has a degree-zero certificate.

| Quantity | Meaning |
|---|---|
| $T$ | Number of Toffoli gates; NOT and CNOT have zero cost in this quantity |
| $L$ | Total number of NOT, CNOT, and Toffoli gates |
| $M$ | Number of binary AND operations in an XOR–AND computation |
| $n$ | Actual number of input bits, including all variables in a displayed family |
| $\dim_K A$ | Dimension of a specified linear summary algebra over $K$ |

A bound on one row of this table cannot silently be substituted for a bound on another row. In particular, a computation can have few Toffolis and many CNOTs. An exponentially large ambient algebra can have succinct expressions. Those distinctions explain the projection and gluing examples later.

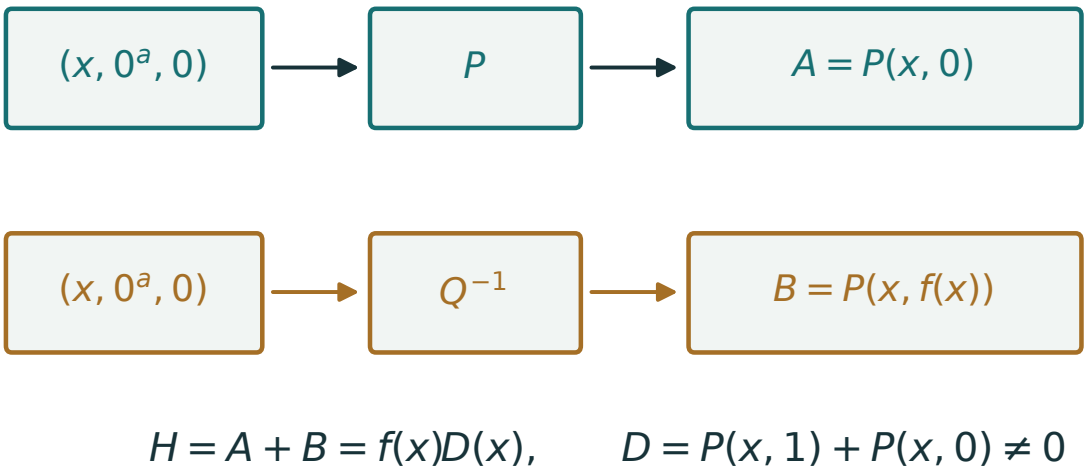
**Source.** `docs/FOUNDATIONS.md,Conventions;formalization/lean/CircuitObstructions/Gates.lean`

## An arbitrary cut compares two reversible computations

Write the circuit as $C = Q \circ P$. The prefix $P$ runs forward; the inverse suffix $Q^{-1}$ starts from the same zero-target boundary state. Define

$$A(x) = P(x,0), \qquad B(x) = Q^{-1}(x,0), \qquad D(x) = P(x,1) + P(x,0).$$

The zero ancillas are suppressed here. The two executions generally arrive at different interior states, even though they start at matching boundaries.




The exact C01 objects. Both arrows remain inside the legal reversible state space.

Correctness on both target branches gives

$$Q^{-1}(x,b) = P(x,b + f(x)).$$

To derive it, apply $C$ to $(x, b + f(x))$ and then apply $Q^{-1}$ to the result. For $b = 0$, $B = A$ when $f(x) = 0$, while $B = P(x,1)$ when $f(x) = 1$. Thus

$$H(x) := A(x) + B(x) = f(x)D(x).$$

The two original states differ in their target. Since $P$ is injective, $D(x) \neq 0$. Each coordinate discrepancy is supported on positives, and some coordinate discrepancy detects every positive input.

**Evidence.** General written identity; archived formal declarations `cut_transport_identity`, `direction_nonzero`, and `complete_safe_cover`.

## Why completeness is pointwise, and why no decoder is needed

For a coordinate $i$, let $H_i = A_i + B_i$. The cut identity says

$$H_i(x) = f(x)D_i(x), \qquad H_i \le f, \qquad \bigvee_i H_i = f.$$

The last equation is pointwise. For every fixed accepted $x$, at least one coordinate has $D_i(x) = 1$. The detecting coordinate may depend on $x$ and may be an input, scratch, or target wire. The argument does not require a single fixed coordinate to detect all positives.

At an empty prefix, the direction is simply the target unit vector. At a full prefix, all non-target wires have restored and the target discrepancy is $f$. Interior cuts can spread the direction through the workspace. These minimum cases verify the composition order and the role of the two target branches.

The zero-input cases are also valid. With only a target wire, the empty circuit realizes the constant-zero oracle and one NOT realizes the constant-one oracle. At every cut, the direction is nonzero and the discrepancy cover has the correct accepting set.

> **Essential hypothesis.** The endpoint condition holds for both initial target values. The inverse-suffix derivation uses the branch $b + f(x)$. Correctness only for initial target zero is not the stated model.

The theorem also does not require scratch to remain zero at the cut. Imposing that extra condition would exclude admitted restored circuits. Conversely, dropping target separation from the boundary encoding permits a vacuous endpoint equation: an encoding that ignores $y$ can make the empty circuit appear to compute a nonzero function. The formal `Boundary.target_eq` property prevents precisely this degeneracy.

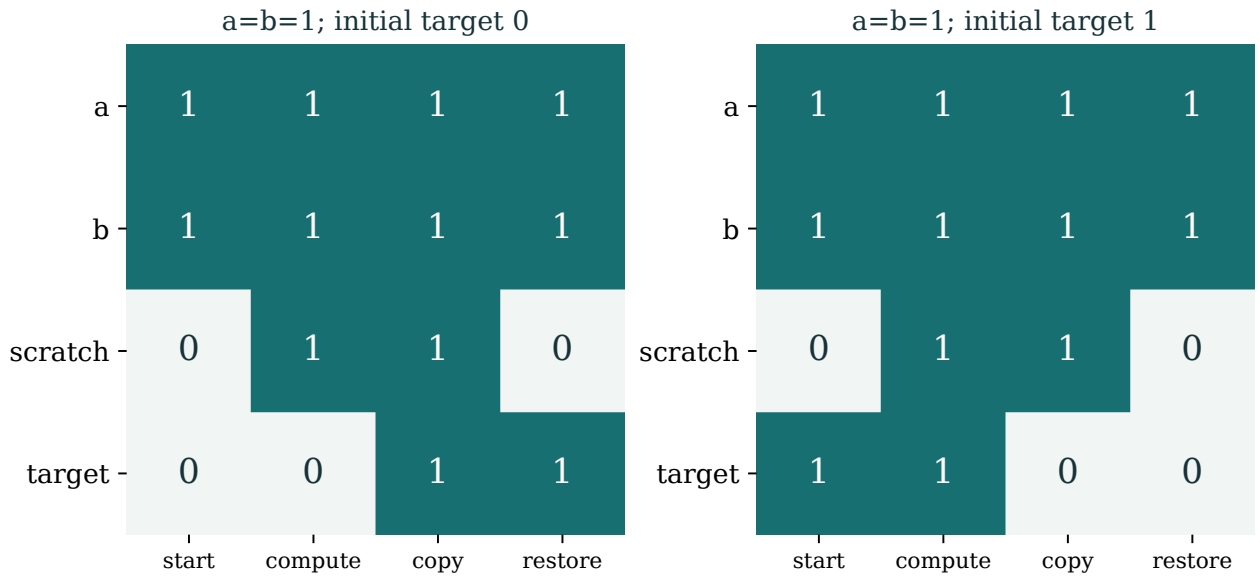
**Source.** `CutComparison.lean`:
`inverse_suffix_boundary`, `every_index_cut`, `canonical_complete_safe_cover`

## A dirty scratch wire can be fully restored

Consider the four-wire oracle in the Lean examples. First compute $ab$ into scratch $s$, copy $s$ into the target, and uncompute $s$:

$$(a,b,s,y) \xrightarrow{T(a,b;s)} (a,b,s+ab,y) \xrightarrow{C(s;y)} (a,b,s+ab,y+s+ab) \xrightarrow{T(a,b;s)} (a,b,s,y+s+ab).$$

The displayed final expression specializes to the required oracle when the boundary scratch is zero. On the positive input $a=b=1$, the scratch is dirty between the first and last gates.




Exact states for the accepting input, in wire order $(a,b,s,y)$. White cells are zero; teal cells are one.

The two panels check both initial targets. The final target is toggled, the logical inputs are unchanged, and scratch returns to zero. A proof that used an intermediate clean-workspace assumption would fail to cover this explicit legal fixture.

This circuit uses two Toffolis. It is a useful restoration witness, not the optimal two-input AND oracle: the direct gate $T(a,b;y)$ computes that oracle with one Toffoli and no scratch. The guide distinguishes an exact example from an optimization claim.

**Evidence.** Exact new trace assertion imported from the source gate semantics; the same fixture is formalized by `Examples.andOracle_restores`.

## The output direction can move away from the target

Two data/target wires already suffice. A SWAP is implemented by three legal CNOTs. Let $P$ be that SWAP, and let $Q$ be NOT on the data wire followed by another SWAP. Then

$$(x,y) \xrightarrow{P} (y,x) \xrightarrow{Q} (x,1+y).$$

The whole circuit is a restored constant-one oracle with zero Toffolis. At the cut after the first SWAP,

$$A = P(x,0) = (0,x), \qquad B = Q^{-1}(x,0) = (1,x), \qquad H = (1,0).$$

The target discrepancy is zero on every input, while the data discrepancy is one on every input. Reading only the target at this cut would reject the constant-one function.

| $x$ | $A$ | $B$ | $H$ | named target discrepancy |
|---|---|---|---|---|
| 0 | $(0,0)$ | $(1,0)$ | $(1,0)$ | 0 |
| 1 | $(0,1)$ | $(1,1)$ | $(1,0)$ | 0 |

This example uses the target as a control during each SWAP. That behavior is expressly allowed by the model. Its relevance is logical: completeness belongs to the full register comparison, not to an arbitrarily chosen wire. The coordinate packets remain perfectly safe even though the detection direction has moved.

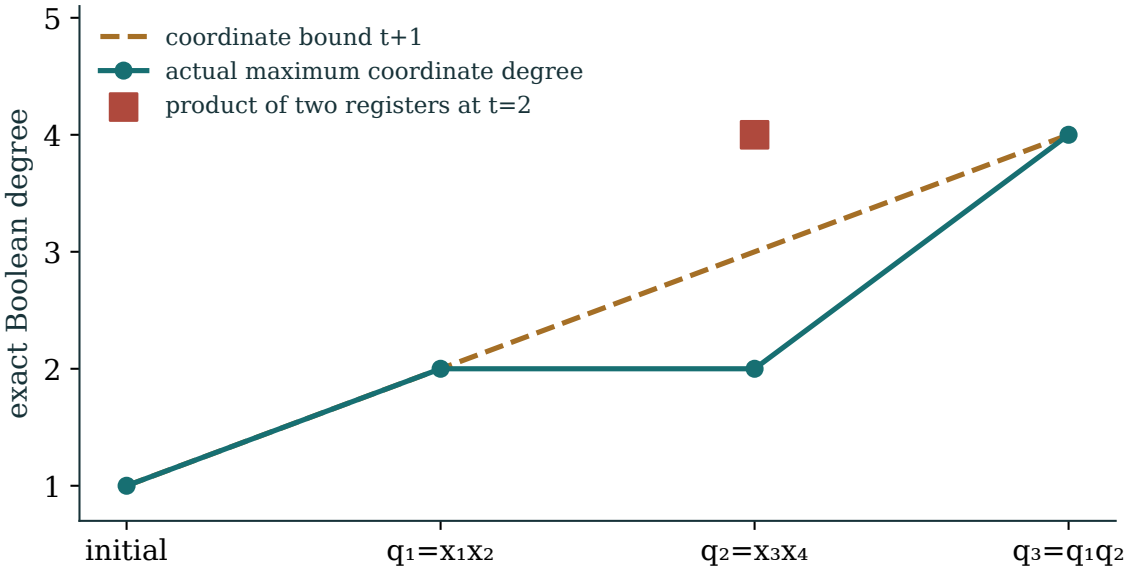
**Evidence.** Constructive formal counterexample to target-only reconstruction: `outputActiveOracle_restores`, `moved_discrepancy_data`, `moved_discrepancy_target`.

## Degree bounds must retain shared register products

Take four independent input bits. Compute

$$q_1 = x_1x_2, \qquad q_2 = x_3x_4, \qquad q_3 = q_1q_2.$$

After two ANDs, each nonlinear register has degree two, but their product has degree four. After the third AND, a coordinate of degree four appears.




Exact degrees from the complete four-input truth tables. The square marker is a two-register product, not a coordinate.

The correct invariant tracks a product of $r$ current registers after $t$ Toffolis:

$$\deg\left(\prod_{i\in L} s_i(x)\right) \le t + |L| = t + r.$$

At $t = 2$, the product $q_1q_2$ has $r = 2$ and saturates $t + r = 4$. At $t = 3$, the resulting coordinate has $r = 1$ and saturates $t + 1 = 4$.

This explains why tracking only individual register degrees and then adding their separate histories can lose the shared accounting. The invariant applies to the actual evolving state. It does not unfold a reused register into a new copy of the entire computation.

**Evidence.** Exact finite fixture plus the general register-product invariant; the source proves `workspaceDegree_run` for arbitrary gate lists.

## The invariant handles duplicates, NOT, and CNOT

Suppose a gate toggles target $u$ by a product of $c$ controls. In a product list $L$ that contains $u$, remove all occurrences of $u$ to obtain $R$. Boolean idempotence leaves one updated target factor, and

$$\prod_{i\in L} s_i' = s_u \prod_{i\in R} s_i + \left(\prod_{j\in \text{controls}} s_j\right) \prod_{i\in R} s_i.$$

The old invariant bounds these terms by $t+1+|R|$ and $t+c+|R|$. Since $|R|+1 \le |L|$, the budget rises by at most $\max\{c-1, 0\}$.

| Gate | $c$ | toggle expression | extra budget |
|---|---|---|---|
| NOT | 0 | 1 | 0 |
| CNOT | 1 | one control | 0 |
| Toffoli | 2 | two controls | 1 |

If the target is absent from $L$, the product is unchanged. An empty list is the constant-one product. Repeated target indices are removed using idempotence; repeated other indices can only reduce multilinear degree. Controls already appearing in $R$ also cause no difficulty.

For general XOR–AND computations, the written proof introduces a Boolean variable for each AND output. Every wire is affine in inputs and preceding gate variables. Substitute those variables in reverse topological order. A squarefree monomial contains each substituted variable at most once, so replacing it by a quadratic raises degree by at most one. After $t$ substitutions, the affine output has degree at most $t+1$.

These are two precise proof routes. The general syntax argument is written; the maintained NCT register argument is in Lean. The formalization does not assert a separate arbitrary XOR–AND compiler or a canonical ANF algorithm.

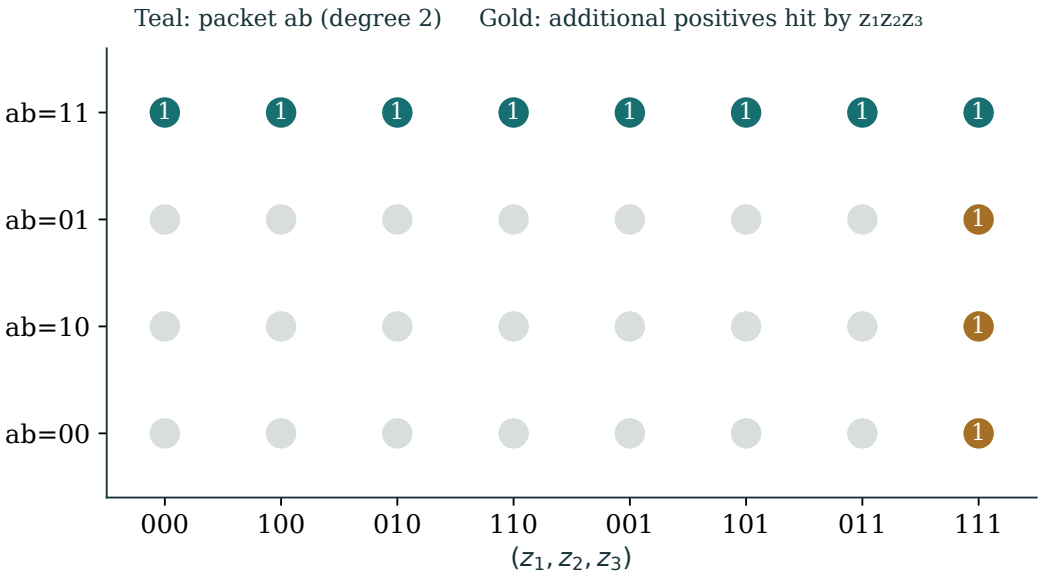
**Source.** `MultiplicativeDegree.lean;docs/FOUNDATIONS.md,F3`

## One safe packet need not cover every accepting point

For $k = 3$ consider

$$F_3(a, b, z_1, z_2, z_3) = ab \text{ OR } z_1z_2z_3.$$

The packet $ab$ is safe and has degree two. It detects every teal point below. It misses three additional positives, which require the other packet $z_1z_2z_3$.




All 32 assignments. There are 11 positives: 8 in the teal row and 3 additional gold points. The coordinate labels use $(z_1, z_2, z_3)$ order.

The one-packet parameter $\alpha_1$ asks for any nonzero safe polynomial. The complete parameter $\beta$ asks for safe detection of every positive:

$$\beta_v(f) = \min\{\deg h : h \le f,\ h(v) = 1\}, \qquad \beta(f) = \max_{v:f(v)=1} \beta_v(f).$$

For this monotone function, its minimal positive supports have sizes two and three. Thus $\alpha_1(F_3) = 2$ while $\beta(F_3) = 3$. Selecting one easy accepting region loses the pointwise obstruction at the other region.

**Evidence.** All 32 points and the two parameter values were freshly recomputed; the general monotone minterm characterization is a written proof for nonzero functions.

## Balanced cuts turn coverage into a Toffoli bound

Choose a cut after $\lfloor T/2 \rfloor$ Toffolis. Both sides have at most $\lceil T/2 \rceil$ Toffolis. Reversing the suffix preserves its count because every gate is self-inverse. Each forward or inverse-suffix coordinate has degree at most $\lceil T/2 \rceil + 1$, and their XOR has the same upper bound.

For every positive $v$, completeness chooses a discrepancy $H_i$ that detects $v$. Its safety and degree bound imply

$$\beta(f) \le \left\lceil \frac{T}{2} \right\rceil + 1, \qquad 2\beta(f) \le T + 3.$$

The integer rounding matters. If $T = 2r$, the first bound gives $2\beta \le T + 2$. If $T = 2r + 1$, it gives $2\beta \le T + 3$. The uniform bound uses the latter constant.

An affine output mask $\ell(x)$ costs zero additional Toffolis: append CNOTs from restored inputs and, if needed, a target NOT. This constructs a restored oracle for $f + \ell$, so the bound applies to every affine mask. Define

$$\beta^*(f) = \max_{\ell \text{ affine}} \beta(f + \ell).$$

Together with the output degree bound,

$$\boxed{T \ge \max\{0, \deg f - 1, 2\beta^*(f) - 3\}.}$$

There are finitely many affine masks. The Lean definition uses the least upper bound of their beta values, which equals this finite maximum. Its affine-output theorem constructs legal gates rather than assuming masks are free transformations of an arbitrary interior computation.

**Evidence.** Written theorem and archived endpoint `restored_degree_and_cover_bound`; the dual is a separate written result.

## An affine shift reveals a harder accepting point

For $k \geq 2$, let $B = z_1 \cdots z_k$ and $F_k = ab \text{ OR } B$ on $n = k+2$ inputs. Add the affine mask $a$ to obtain $G_k = F_k + a$. On the face $b = 1$,

$$G_k|_{b=1} = (1+a)B.$$

This is a singleton indicator on the $k+1$ variables $(a, z)$. At $v = (a = 0, b = 1, z = 1^k)$, every safe detecting polynomial must restrict to that singleton indicator, so its degree is at least $k+1$.

The general two-positive indicator construction gives the matching upper bound $\beta(G_k) \leq n-1 = k+1$. The maintained family proves

$$\max_{\ell} \alpha_1(F_k + \ell) = 3, \qquad \beta^*(F_k) = k+1.$$

The gap between one-packet and complete degree therefore grows with $k$.

| $k$ | $n$ | $\beta^*$ | complete bound $2k-1$ | combined degree/one-packet bound |
|---|---|---|---|---|
| 2 | 4 | 3 | 3 | 3 |
| 3 | 5 | 4 | 5 | 4 |
| 4 | 6 | 5 | 7 | 5 |
| 6 | 8 | 7 | 11 | 7 |

For $k = 3$, all 64 affine masks were freshly checked and the optimized parameters are exactly $(3, 4)$. The all-size family proof is not inferred from those 64 cases. Its restored construction has $T \leq 2n-3$, so this remains a linear-size easy family. The exact optimum between the lower and upper constructions is not asserted when they differ.

**Source.** `docs/THEORY.md,section2;reviews/visual_examples.json,unbalanced_or`

## A degree-one dual certificate for two-input AND

Let $f(a,b) = ab$, with unique positive point $(1,1)$. For degree at most one, the evaluation vector is $v_1(a,b) = (1,a,b)$. The three negative points already span the positive vector.

| point (a,b) | 1 | a | b | AND(a,b) |
|---|---|---|---|---|
| (0, 0) | 1 | 0 | 0 | 0 |
| (1, 0) | 1 | 1 | 0 | 0 |
| (0, 1) | 1 | 0 | 1 | 0 |
| (1, 1) | 1 | 1 | 1 | 1 |

$$v_1(11) = v_1(00) + v_1(10) + v_1(01)$$

The first three rows are negative inputs. Their XOR equals the last evaluation row.

Any affine polynomial is a coefficient covector $c = (c_0, c_a, c_b)$. Safety means that it pairs to zero with each negative row. The displayed dependence therefore forces its value at $(1,1)$ to be zero. No safe affine packet detects the positive point.

The degree-two polynomial $ab$ does detect it, so $\beta_{11}(f) = \beta(f) = 2$. Since $\beta^* \le n = 2$ and $\beta^* \ge \beta$, $\beta^* = 2$. The complete bound gives $T \ge 1$, attained by one direct Toffoli to the target.

The constant coordinate also checks parity: this dependence uses three negative rows, an odd number. Without the constant coordinate, that parity conclusion would not follow.

**Evidence.** Exact finite dependence plus written duality. Lean proves the corresponding AND2 degree and beta exclusions by four-corner parity, without formalizing the general evaluation-span dual.

## The finite evaluation-span dual in full

Fix a positive $v$ and integer $d \geq 0$. Let

$$v_d(x) = (x^S)_{S\subseteq[n],\ |S|\leq d}, \qquad U = f^{-1}(0), \qquad N = \operatorname{span}_{\mathbb{F}_2}\{v_d(u) : u \in U\}.$$

The constant monomial $S = \varnothing$ is included. Degree-$d$ coefficient covectors form the finite dual of this evaluation space. A covector is safe exactly when it annihilates $N$.

If $v_d(v) \in N$, every safe covector vanishes at $v$. If $v_d(v) \notin N$, extend a basis of $N$ by $v_d(v)$ and then to a basis of the full space. Assign zero to the $N$ basis and one to $v_d(v)$. The resulting covector is a safe detector. Hence

$$\boxed{\beta_v(f) > d \quad \Longleftrightarrow \quad v_d(v) \in N.}$$

Choose a basis of $N$ from negative evaluation rows. A span representation of the positive row gives a subset $W \subseteq U$ with

$$v_d(v) = \sum_{u\in W} v_d(u), \quad |W| \text{ odd}, \quad |W| \leq \sum_{j=0}^{\min(d,n)} \binom{n}{j}.$$

It is a replayable finite obstruction to every safe degree-$d$ detector.

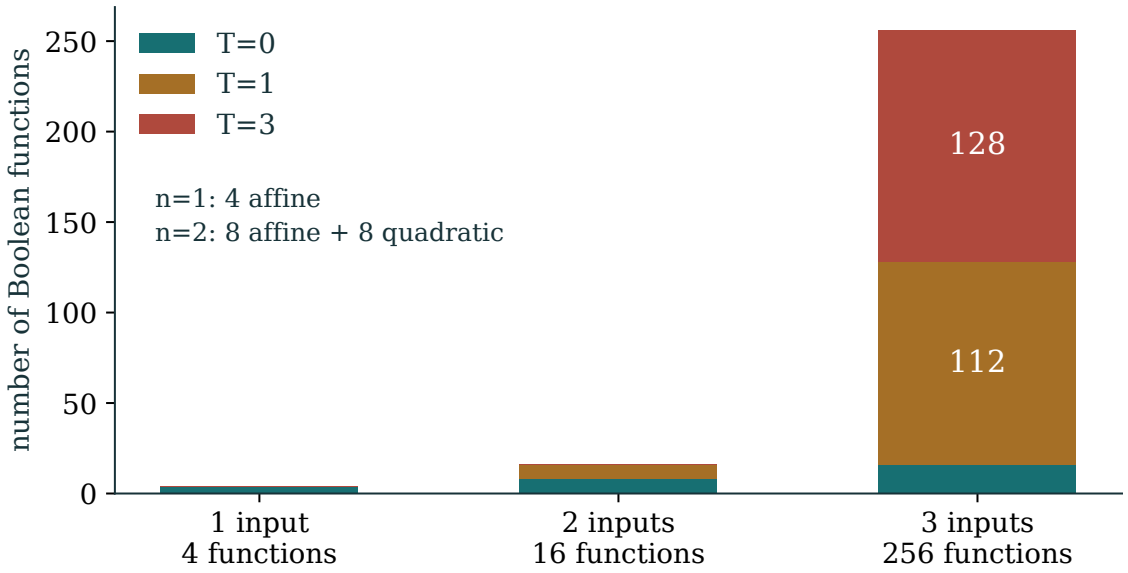
At $d = 0$, a negative exists exactly when the constant-one detector is unsafe. At $d \geq n$, the full evaluation rows are independent and no positive row lies in the negative span. This matches $\beta_v \leq n$ from point indicators. For $f = 0$, there is no positive $v$ and the separate convention is $\beta(0) = 0$.

Fresh enumeration checked all 278 functions on zero through three inputs, all positives, and every $d = 0, \ldots, n$: 4,201 primal/dual checks agree. Those bounded checks support the independently supplied universal proof.

**Source.** `scripts/core_dual_checks.py;reviews/core_dual_checks.json`

## Every function through three inputs has a matching oracle

The small calibration concerns every truth table on one, two, or three inputs. It is not a search through all reversible circuit words. A theorem lower bound and an explicit construction coincide.




Reconstructed source constructions for all 276 functions. Totals are 28 with $T = 0$, 120 with $T = 1$, and 128 with $T = 3$.

Affine functions use target CNOTs and possibly NOT. A nonaffine quadratic through three bits has alternating quadratic rank two, so it is a product of two independent linear forms plus an affine residual. Compute those forms on clean wires, apply one Toffoli, uncompute, and append the residual.

Every three-bit cubic is a translated three-variable monomial plus an affine residual. The singleton monomial needs three Toffolis by complete coverage, and compute/toggle/uncompute attains that number.

The new figure script rebuilds and validates all 276 circuits and both target branches, covering 4,240 scalar input/target cases via exact truth tables. Its totals match the frozen baseline. Zero-input constants were checked in the separate dual suite, rather than being added to this 276-function count.

**Evidence.** Matching written lower bounds and exact constructed implementations; not exhaustive optimal-circuit synthesis. The generic monomial cost theorem is not newly Lean-certified by this edition.

## Projection is defined in the original coordinates

For $m \geq 2$, put $a = (1, \ldots, 1)$ and

$$V = \mathbb{F}_2^m \setminus \{0, a\}, \qquad n = 2^m - 2.$$

There is one input bit $x_v$ for each $v \in V$. Define

$$L_j(x) = \sum_{v : v_j = 1} x_v, \qquad P_m(x) = \prod_{j=1}^{m} L_j(x).$$

The linear map $x \mapsto L(x)$ has rank $m$, because the standard basis columns are present. In the $L$ coordinates, $P_m$ is a monomial in $m$ independent bits. Its restored nonlinear cost is $2m - 3$.

Now extract the quadratic homogeneous ANF component $Q_m$ in the *original* $x_v$ coordinates. Since $P_m(0) = P_m(e_v) = 0$, for distinct $u, v$ its quadratic coefficient is

$$[x_u x_v] P_m = P_m(e_u + e_v) = [u + v = a].$$

The involution $v \mapsto v + a$ pairs every variable with one other variable:

$$Q_m = \sum_{\{v, v+a\}} x_v x_{v+a}.$$

It is a quadratic inner product on $n/2$ disjoint pairs. Its alternating polar matrix has full rank $n$.

The coordinate choice is essential. Homogeneous grade is not invariant under arbitrary affine changes of variables. A cheap representation in the $L$ coordinates does not imply cheap independent generation of the quadratic grade in the original coordinates. The comparison keeps the definition of projection fixed throughout.

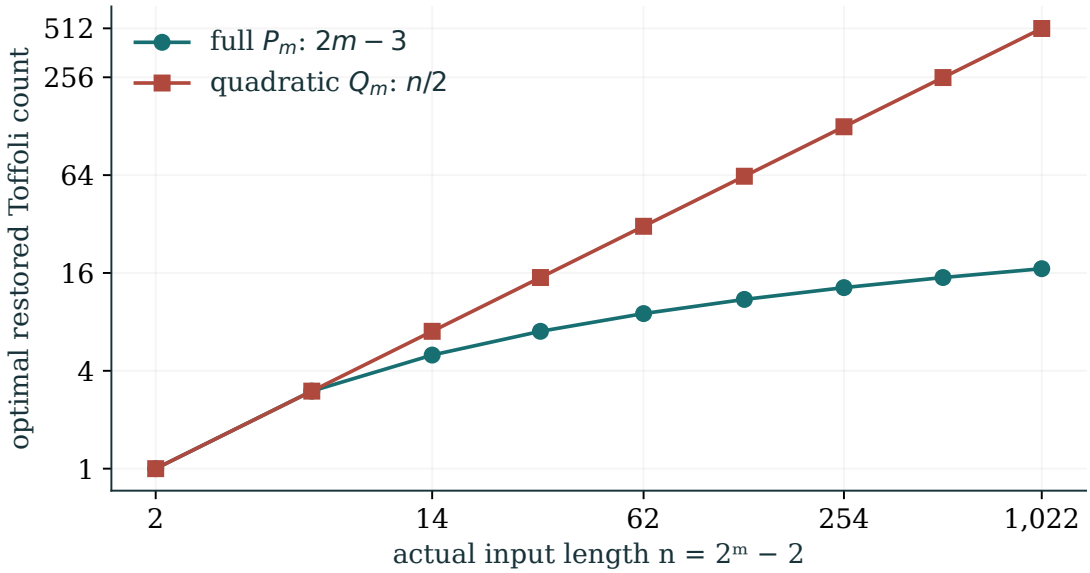
**Source.** `docs/THEORY.md,section3;checks/verify_projection.py`

## Exact nonlinear projection costs versus actual input length

The source proves the all-size formulas

$$M(P_m) = m - 1, \quad T_{\text{rest}}(P_m) = 2m - 3, \qquad M(Q_m) = T_{\text{rest}}(Q_m) = n/2.$$

The chart uses actual input length $n = 2^m - 2$ on its horizontal axis.




Exact formulas for $m = 2, \ldots, 10$. Both axes are logarithmic; the plotted points are not optimal-circuit search results.

For $Q_m$, each disjoint quadratic pair is accumulated directly into the target with one Toffoli. The lower bound allows nonlinear intermediates: any computation with $t$ multiplications is affine on an affine subspace of codimension at most $t$. A nondegenerate quadratic can be affine only on a subspace with isotropic direction space, whose dimension is at most $n/2$. Thus $t \geq n/2$.

The separation grows exponentially in $m$, but $n$ itself grows exponentially in $m$. The quadratic cost is linear in actual input length. The conclusion is about nonlinear generation and a specified grade extraction operation, not a superpolynomial unrestricted circuit bound.

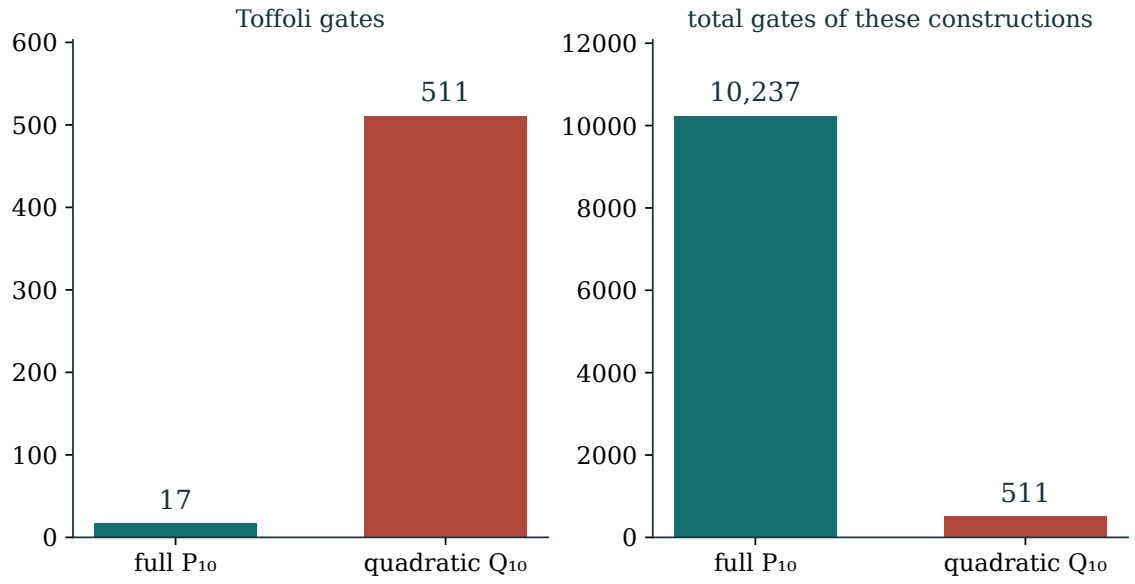
**Evidence.** All-size written cost proofs; constructor gate counts freshly asserted through $m = 10$. No full truth-table enumeration at $m > 4$ is claimed.

## At $m = 10$, the total-gate accounting points the other way

Here $n = 1,022$. The full function uses 17 Toffolis; its quadratic grade needs 511. However, computing and uncomputing the $m$ parities also uses

$$2m(2^{m-1} - 1) = 20 \cdot 511 = 10,220 \quad \text{CNOTs.}$$

The explicit full construction has $10,237$ total gates. The quadratic accumulation uses only 511 total gates.




Two different resource comparisons at the same input length. The right panel reports the totals of the stated constructions, not an asserted optimal total for $P_{10}$.

The left panel is a valid nonlinear-cost separation. The right panel prevents interpreting it as an exponential ordinary total-gate separation. The total full construction is $O(n \log n)$, whereas the quadratic construction is $n/2$ gates.

A proposed invariant that charges an original computation for every separately extracted grade would have to justify that extraction's cost. These formulas refute an overhead bounded only by a polynomial in the original nonlinear count, independently of input and linear resources. They do not exclude an overhead polynomial in $n + T$ or a different precisely defined invariant.

**Evidence.** Exact counts obtained from the source constructors; optimal Toffoli formulas have written proofs. The full total-gate count is an upper construction count.

## Quadratic projection can destroy safety

At $m = 3$, variables are ordered by source labels $v = 1, 2, 3, 4, 5, 6$. Assignment index 13 is read least-significant-bit first:

$$(x_1, x_2, x_3, x_4, x_5, x_6) = (1, 0, 1, 1, 0, 0).$$

The active labels have syndrome $1 + 3 + 4 = 6$, not $a = 7$.

| input | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $x_6$ |
|---|---|---|---|---|---|---|
| label v | 1 | 2 | 3 | 4 | 5 | 6 |
| x (13) | 1 | 0 | 1 | 1 | 0 | 0 |

$$L = (0, 1, 1) \Rightarrow P_3 = 0; \quad Q_3 = x_1x_6 + x_2x_5 + x_3x_4 = 1$$

A concrete accepted point for $Q_3$ that is rejected by $P_3$. The variable ordering matches the source checker.

The three linear forms evaluate to $L = (0, 1, 1)$, so $P_3 = L_1L_2L_3 = 0$. The quadratic grade has pairs $(1, 6)$, $(2, 5)$, $(3, 4)$, so

$$Q_3 = x_1x_6 + x_2x_5 + x_3x_4 = 0 + 0 + 1 = 1.$$

Thus $Q_3 \notin I_{P_3}$: homogeneous extraction does not preserve the positive-support ideal. The whole polynomial is supported on its own accepting set, but its isolated homogeneous components need not be.

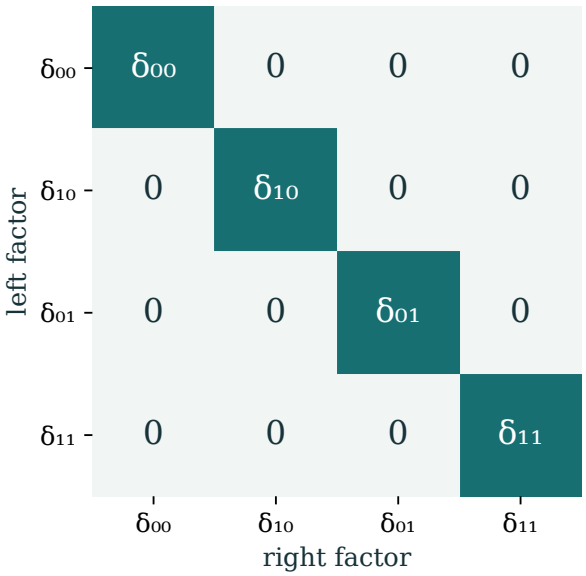
The new script checks all 64 inputs here, finds 24 false positives, and confirms that 13 is the first in the stated integer ordering. The frozen $m = 4$ control similarly has 7,616 false positives among its $2^{14}$ assignments. These are separate finite statements with exact ranges.

**Evidence.** Fresh complete $m = 3$ truth-table check; baseline $m = 4$ check was reproduced by the full source verifier. Safety failure is already established by the single valid assignment above.

## Witness indicators are orthogonal idempotents

Let $K$ be a field and $W = \{0,1\}^m$. The full witness algebra is $B = K^W$ with pointwise addition and multiplication. For each witness $v$, the indicator $\delta_v$ is one at $v$ and zero elsewhere. Then

$$\delta_v^2 = \delta_v, \qquad \delta_u \delta_v = 0 \ (u \neq v), \qquad \sum_v \delta_v = 1.$$




The four-witness multiplication table at $m = 2$. Each nonzero diagonal entry is its own point indicator.

The four indicator functions form a basis of $K^4$. General $m$ has $2^m$ such basis elements. A verifier is the sum of the indicators of its accepted witnesses, and pointwise multiplication models intersection of accepting sets.

This algebra retains exact witness compatibility: disjoint singleton predicates multiply to zero. A summary that preserves that product operation while retaining every singleton must preserve the same orthogonal structure. The following dimension theorem says exactly how much fixed linear space that requires.

**Evidence.** Exact $m = 2$ multiplication table; general product-algebra identities hold over any field. No characteristic-two assumption is needed for the dimension proof.

## Faithful linear multiplicative summaries need $2^m$ dimensions

Suppose $\Phi : K^W \to A$ is $K$-linear and multiplicative, and $\Phi(\delta_v) \neq 0$ for every $v \in W$. Put $e_v = \Phi(\delta_v)$. The images are nonzero orthogonal idempotents.

If $\sum_v c_v e_v = 0$, multiply by $e_u$. Orthogonality removes every other term and idempotence gives $c_u e_u = 0$. Since $K$ is a field and $e_u \neq 0$, $c_u = 0$. Hence all $e_v$ are linearly independent and

$$\boxed{\dim_K A \geq |W| = 2^m.}$$

No unital hypothesis is needed. At $m = 0$, there is one empty witness and the bound is simply $\dim_K A \geq 1$.

For a unital scalar map to $K$, scalar idempotents are zero or one. Orthogonality permits at most one nonzero image, and preservation of one requires exactly one. Such a map is evaluation at one witness. For $m \geq 1$ it loses at least one other singleton verifier. For $m = 0$, there is no other witness to lose.

> **Model boundary.** The theorem quantifies a fixed summary serving all predicates at a witness length, preserving linear combinations and exact pointwise multiplication. An unrestricted algorithm need not use such a summary. Syntax, nonlinear encodings, changing spaces, context-dependent operations, and succinct expressions are outside these hypotheses.

An exponentially large algebra can still contain short circuit expressions. Therefore exponential dimension does not imply exponential expression length or a lower bound for every algorithm on one explicit decision family.

**Source.** `docs/THEORY.md,Theorem4.1;reviews/extended_audit.md`

## Distinct witness monomials prevent formal cancellation

Over a characteristic-two field, define

$$P_R(z) = \sum_{w \in \{0,1\}^m} R(w) \prod_i z_i^{w_i}.$$

The $z_i$ are formal indeterminates, not Boolean variables. Different witnesses produce different monomials, so $P_R = 0$ exactly when there is no accepting witness.

| witness ($w_1$,$w_2$,$w_3$) | R(w) | retained monomial |
|---|---|---|
| 000 | 0 | — |
| 100 | 1 | z1 |
| 010 | 1 | z2 |
| 110 | 0 | — |
| 001 | 0 | — |
| 101 | 1 | z1z3 |
| 011 | 1 | z2z3 |
| 111 | 0 | — |

An exact three-bit verifier accepts witness indices $1, 2, 5, 6$ in least-significant-bit order. Four accepts have parity zero, but their monomials are distinct.

Here

$$P_R = z_1 + z_2 + z_1z_3 + z_2z_3 = (z_1 + z_2)(1 + z_3) \neq 0.$$

The ordinary unweighted characteristic-two sum of the four accepts is zero. The formal polynomial avoids this cancellation by retaining witness labels as monomial supports.

Nonzero as a formal polynomial does not mean nonzero at every field point. For instance, this example vanishes whenever $z_1 = z_2$ or $z_3 = 1$. Random evaluation is governed by a root bound, not by the formal identity alone. Evaluating the polynomial efficiently remains a separate task.

**Evidence.** Exact support fixture; general no-cancellation argument is written. The source finite evaluator retains $2^m$ coefficients.

## A short expression inside a trace is not a computed trace

Let $\tau_w$ extract the top multilinear coefficient in the Boolean witness ring. For any $G(w)$ over a characteristic-two field,

$$\tau_w(G) = \sum_{v \in \{0,1\}^m} G(v).$$

A proper-support monomial evaluates to one an even number of times; the full-support monomial does so once. This proves the identity directly.

Applying it to a weighted expression gives

$$P_R(z) = \tau_w \left[ R(w) \prod_{i=1}^{m} (1 + (1 + z_i) w_i) \right].$$

At $w_i = 0$, a multiplier equals one. At $w_i = 1$, it equals $z_i$. Summing evaluations therefore produces exactly the witness monomials. For the preceding verifier, the four surviving products are $z_1, z_2, z_1 z_3, z_2 z_3$.

The circuit expression *inside* the trace adds only $O(m)$ operations to the verifier. That observation bounds the expression's construction, not the cost of applying $\tau_w$. The finite checker computes with a complete table of $2^m$ witness coefficients.

| Established | Not supplied by the expression |
|---|---|
| Small weighted circuit inside $\tau$ | A polynomial-time evaluator for the trace |
| Formal polynomial detects nonemptiness | A deterministic identity test from one evaluation |
| Exact finite table algorithm | Polynomial storage in the witness length |

At $m = 0$, the empty product is one, the top coefficient is the constant itself, and there is a single empty witness. The definitions remain valid.

**Source.** `docs/THEORY.md, equations (10)--(11); checks/verify_fingerprint.py`

## Disjoint-variable factorization uses each predicate's own block

Suppose $A$ depends on witness variables in block $S$ and $B$ on a disjoint block $T$. Write $\tau_{z_S}(A) = \sum_s A(s) z_S^s$ for the weighted trace on $S$, distinct from the unweighted top-coefficient notation $\tau_w$. Weighted sum over the union factors as

$$\sum_{s,t} A(s)B(t) z_S^s z_T^t = \left(\sum_s A(s) z_S^s\right)\left(\sum_t B(t) z_T^t\right).$$

Each factor is traced on its own variable block. The union trace is then their product. This is an ordinary factorization of a finite double sum.

For $A = w_1$ and $B = w_2$,

$$\tau_{z_1}(A) = z_1, \qquad \tau_{z_2}(B) = z_2, \qquad \tau_{z_1,z_2}(AB) = z_1 z_2.$$

Tracing both factors over the full two-bit ambient domain introduces unused variable sums:

$$\tau_{z_1,z_2}(A) = z_1(1+z_2), \qquad \tau_{z_1,z_2}(B) = z_2(1+z_1).$$

Their product generally differs from the union trace. At $z_1 = z_2 = 1$, each full ambient factor is zero, while the trace of $AB$ is one. Disjoint dependence alone does not justify multiplying traces with mismatched domains.

The source checker tests all four one-bit predicates per block and every pair of GF(16) weights: $4 \cdot 4 \cdot 16^2 = 4{,}096$ own-block factorization cases. It does not claim an ambient-domain product identity.

**Evidence.** General written factorization under the own-block convention; exact finite source check and an explicit ambient-domain counterexample.

## Overlap breaks the attempted context-free product rule

One shared witness bit already defeats multiplicativity. With $\tau_z(P) = P(0) + zP(1)$,

$$\tau_z(w) = z, \qquad \tau_z(1+w) = 1, \qquad \tau_z(w(1+w)) = 0.$$

The product of the separate summaries is $z$. Also $w^2 = w$ in the Boolean ring, so $\tau_z(w^2) = z$, whereas $\tau_z(w)^2 = z^2$.

| expression | trace after product | product of traces |
|---|---|---|
| w(1+w) | 0 | θ |
| $w^2 = w$ | θ | $\theta^2$ |

GF(16) = $F_2$[θ]/($\theta^4$+θ+1); θ = integer 2, $\theta^2$ = integer 4

A concrete field point $z = \theta$ with $\theta \neq 0, 1$. The two columns are unequal in both rows.

The field is $\mathbb{F}_{16} = \mathbb{F}_2[\theta]/(\theta^4 + \theta + 1)$. The checker stores field elements as four-bit coefficient vectors: $\theta$ is integer 2 and $\theta^2$ is integer 4. These integers encode field elements; they are not ordinary integer multiplication. Exact field multiplication gives the reported mismatch.

The first row loses witness compatibility: $w$ and $1 + w$ have no common accepting witness, yet their separate weighted summaries multiply to a nonzero value. The second row loses Boolean idempotence. This is a genuine counterexample to a context-free multiplicative trace compiler, not a floating-point effect or an implementation anomaly.

**Evidence.** Exact GF(16) fixture imported from the source field routines. The general overlap failure is symbolic and does not depend on the selected field.

## Randomized detection is conditional on efficient evaluation

The witness polynomial has total degree at most $m$. Over a finite field $K$, a nonzero polynomial of total degree at most $m$ vanishes on at most an $m/|K|$ fraction of independent uniform field points. The standard inductive proof separates the leading coefficient in the last variable and applies the univariate root bound.

If a uniform polynomial-time exact evaluator for the weighted trace were available, with efficient field representation and arithmetic and $|K| \geq 4m$, one evaluation would accept a satisfiable verifier with probability at least $3/4$. An unsatisfiable verifier has the zero polynomial, so it never produces a false acceptance. Repetition with independent samples reduces the false-rejection probability to at most $4^{-r}$ after $r$ trials.

| Witness bits | GF(16) points | maximum zeros in source tests | degree-bound zeros |
|---:|---:|---:|---:|
| 1 | 16 | 1 | 1 |
| 2 | 256 | 31 | 32 |
| 3 | 4,096 | 721 | 768 |

These source tests enumerate every verifier at each displayed witness length and all field points for the polynomial evaluations. The independent coefficient-extraction comparison has a narrower three-variable grid: coordinates $0, 1, 2, 3$, rather than all GF(16) points. Both scopes are recorded.

This is a conditional randomized algorithmic implication. The required efficient trace evaluator is not supplied. The existing exact evaluator is exponential in witness length, and no deterministic $P = NP$ conclusion is drawn. Random bits and polynomial output description alone do not account for the evaluation operation.

**Source.** `docs/THEORY.md,section5;evidence/baseline/fingerprint.json`

## Reproduce the figures and inspect their exact evidence

All 12 figure pairs are vector PDFs plus raster PNG previews. Their single generation command, from the publication-set root, is

```
python scripts/make_figures.py
```

The script imports the original gate, profile, construction, and GF(16) functions without invoking their release-writing entry points. It writes only new figures and `reviews/visual_examples.json`. That record contains fixture truth-table integers, ordered bits, exact gate lists, scalar counts, the source hashes, and the distinction between finite assertions and plotted all-size formulas.

The separate dual sanity suite runs as

```
python scripts/core_dual_checks.py \
  --output reviews/core_dual_checks.json
```

It compares exhaustive degree-bounded safe truth tables with a separate Gaussian-elimination calculation of the negative evaluation span. Its range is all truth tables at $n = 0, 1, 2, 3$, not an extrapolation.

| Evidence layer | Publication treatment |
|---|---|
| General mathematics | Main paper written proofs with explicit model boundaries |
| Formal subset | Archived Block 02: C01, maintained NCT degree, and numerical C02; no fresh compiler pass |
| New figures | 12 exact generated examples/formula plots, each with fixture provenance |
| Fresh source checks | Complete source `-all` verification and preserved historical outputs passed on 7 October |
| Rendering | This guide is compiled and visually checked separately from mathematical and kernel verification |

The source release, earlier companions, and archived evidence remain unchanged. Publication corrections clarify evidence labels, the nonzero monotone minterm domain, the $m = 0$ scalar-summary boundary, and trace-variable domains. No new conjecture or extension to unrestricted hardness is added.

**Evidence.** Research edition 1.0, 7 October 2026. Common baseline: `komorebi-circuit-obstructions-v1.2.0`. Figure and manuscript sources are included for review and reproduction.